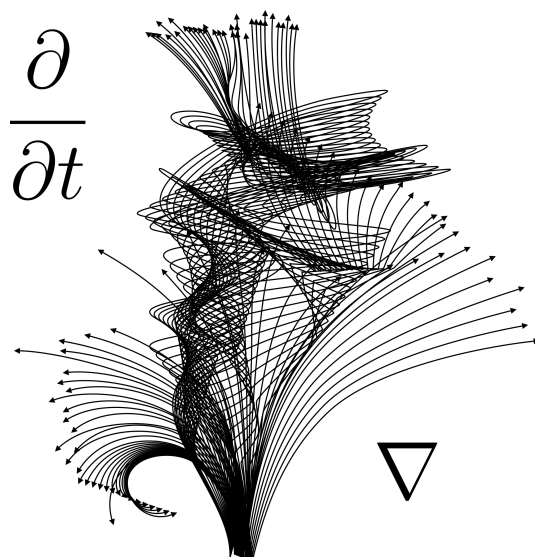




Quantitative Finance Interview Notes

Author: Xiaodong Yang CQF, FRM

Detailed information can be found in the README.md file.

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Chapter 1

Probability, Statistics & Brainteasers

1.1 Probability Foundations (Part A)

Introduction: Why Probability Dominates Quant Interviews

Probability is the single most important skill in quant interviews. A large fraction of questions - brainteasers, logic puzzles, stochastic calculus intuition, derivatives pricing, and risk-neutral reasoning - all reduce to:

- computing expectations
- understanding conditional probability
- analyzing random walks
- reasoning under uncertainty

Firms use probability questions to test your thinking speed, mathematical maturity, and structural reasoning. These problems rarely require memorized formulas; instead, they assess whether you can:

- break a problem into states
- compute conditional paths
- identify symmetry
- reason recursively
- use Markovian structure

This section builds the foundation for the more advanced topics later in the chapter.

Probability Axioms (Interview Level)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

A probability measure satisfies:

1. **Non-negativity:**
 $\mathbb{P}(A) \geq 0$
2. **Normalization:**
 $\mathbb{P}(\Omega) = 1$

3. Countable additivity:

If A_i are disjoint,

$$\mathbb{P}(\cup_i A_i) = \sum_i \mathbb{P}(A_i)$$

Interview expectation:

You never need to formally state these, but many tricky questions rely on:

- breaking events into disjoint pieces
- checking if events are mutually exclusive
- normalizing probabilities

Conditional Probability & Bayes' Rule

Definition

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad (1.1)$$

Bayes' rule

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A) \mathbb{P}(A)}{\mathbb{P}(B)} \quad (1.2)$$

Total probability

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B | A_i) \mathbb{P}(A_i) \quad (1.3)$$

Interview insight: Most conditional probability questions are solved by:

1. Defining states
2. Applying total probability
3. Using Bayes' rule for inversion

Classic Interview Example

The Monty Hall Problem

Example 1.1.1

Setup: You choose 1 of 3 doors; 1 hides a prize. Host opens another door that definitely has no prize. Do you switch?

Solution (interview version):

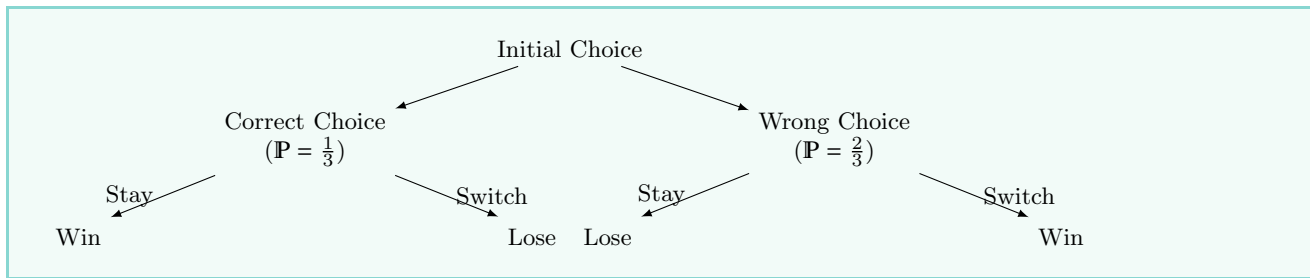
Probability chosen door is correct = $1/3$

Probability prize is behind one of the two unchosen doors = $2/3$

Host removes one losing option

Switching gives probability = $2/3$

Takeaway: If information is not independent (host has intent), conditional probability changes.



Random Variables & Common Distributions

Bernoulli Binary outcome, 0 or 1.

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p) \quad (1.4)$$

Binomial sum of n Bernoullis.

$$\text{mean} = np, \quad \text{variance} = np(1 - p) \quad (1.5)$$

Poisson (arrivals, jumps)

$$\mathbb{P}(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!} \quad (1.6)$$

Key properties:

- Independent increments
- Stationary increments
- Mean = variance = λt

Interview use cases:

- arrival of trades
- number of jumps in jump-diffusion
- queueing logic

Normal Distribution

$$X \sim N(\mu, \sigma^2) \quad (1.7)$$

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \sigma^2 + \mu^2 \quad (1.8)$$

Expectations, Variance & Covariance

Linearity of expectation (critical interview tool):

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad (1.9)$$

Variance:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \quad (1.10)$$

Covariance:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \quad (1.11)$$

Correlation:

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (1.12)$$

Common Quant-Interview Expectation Tricks

Expected value of geometric trials Until first success:

$$\mathbb{E}[N] = \frac{1}{p} \quad (1.13)$$

Examples:

- die rolls until first 6 $\rightarrow 6$
- coin flips until heads $\rightarrow 2$

Expected value of repeated experiments if X_i are IID

$$\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \mathbb{E}[X] \quad (1.14)$$

Symmetry Many brainteasers are solved instantly by symmetry.

Example 1.1.2

Random chord in a circle — probability it is longer than the side of the inscribed triangle?
Answer depends on chosen definition — interviewers check if you question assumptions.

Random Walks (Discrete)

Used in:

- probability puzzles
- gambler's ruin
- Brownian motion intuition
- finance models

1D symmetric walk

$$S_{n+1} = S_n + X_n, \quad X_n = \pm 1 \quad (1.15)$$

Gambler's Ruin Probability you hit upper boundary b before lower boundary 0 starting from i : Symmetric ($p = 0.5$):

$$\mathbb{P}(\text{hit } b) = \frac{i}{b} \quad (1.16)$$

Asymmetric ($p \neq 0.5$):

$$\mathbb{P}(\text{hit } b) = \frac{1 - (q/p)^i}{1 - (q/p)^b} \quad (1.17)$$

This appears frequently in quant interviews.

Markov Chains (Interview Level)

Definition A process X_n is Markov if:

$$\mathbb{P}(X_{n+1} = x | X_n \cdots, X_0) = \mathbb{P}(X_{n+1} = x | X_n) \quad (1.18)$$

Used heavily in:

- queueing
- random walks
- credit rating transitions
- discretized stochastic models

Transition matrix

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \quad (1.19)$$

n-step transitions:

$$p^n$$

Order Statistics (Interview Relevant)

if you draw n IID samples:

- $X_{(1)} = \min$
- $X_{(n)} = \max$

Example 1.1.3

$$\mathbb{E}[\max(U_1, \dots, U_n)] = \frac{n}{n+1} \quad (1.20)$$

Stopping Times (Foundation for Brownian Motion)

A stopping time τ satisfies:

$$\tau \leq t \in \mathcal{F}_t$$

Example 1.1.4

- first time a walk hits $\pm a$
- first time volatility exceeds threshold
- hitting boundary in Brownian motion (early exercise timing)

Interviews rarely ask for formal definition — they want intuition: “**A stopping time is a time at which you stop based only on current information.**”

1.2 Deep Conditional Probability, Classic Puzzles & Core Interview Problems

Conditional Probability — Core Interview Framework

Conditional probability is the single most tested concept in quant interviews. Most puzzles reduce to:

- Identify states
- Assign base probabilities

- Use Bayes' rule
- Compute conditional outcomes
- Interpret intuitively

Below are advanced forms that often confuse candidates.

The Problem of “Useful vs Useless Information”

A recurring theme is distinguishing between:

Independent information

Does not change conditional probability (e.g., rolling a fair die twice)

Dependent information Massively changes the probability (e.g., host showing a goat in Monty Hall)

Key test in interviews:

Always ask yourself: “Does this person KNOW something that influences the outcome?”

Classic Conditional Probability Puzzle Types

Type 1: Revealed information

- Monty Hall
- Cards revealed
- Doors opened
- Host behaviour

Type 2: Random selection vs conditional selection

- “You pick a random child and they are a boy...” vs
- “In a family with at least one boy...”

Type 3: Biased events

- unfair coins
- skewed dice

Type 4: Symmetry + Bayes

- random meeting problems
- birthday paradox
- dart on a square

Interview Classic Problems

Problem 1.2.1 The Two-Child Problem

A family has two children. You meet one of the children randomly, and it is a girl. What is the probability the other child is a girl?

Solution: Incorrect intuition: “Only two possibilities: GB or GG $\Rightarrow 1/2$.”

Correct reasoning: Possible equally likely families: GG, GB, BG, BB

Now you meet a random child, and it's a girl.

Compute probability of observing "Girl met" in each family:

GG family $\rightarrow$ always meet girl (prob = 1)

GB family $\rightarrow$ meet girl w.p. $1/2$

BG family $\rightarrow$ meet girl w.p. $1/2$

BB family $\rightarrow$ never

Weighted probabilities:

GG: 1×1

GB: $(1/2) \times 1$

BG: $(1/2) \times 1$

Normalize:

$$\mathbb{P}(GG|girl) = \frac{1}{1 + \frac{1}{2} + \frac{1}{2}} = \frac{1}{2}$$

Problem 1.2.2 The Drunk Passenger Problem

On a plane with n seats, the first passenger is drunk and picks a random seat. Every subsequent passenger takes their own seat if available; otherwise picks a random available seat. What is the probability the last passenger gets their own seat?

Solution:

Interview solution: This is a symmetry problem.

The last passenger's seat is taken only if:

- the drunk passenger picks their seat first
- OR someone else random-picks it later

The shocking fact:

Once the drunk picks a wrong seat, the problem resets with two "special" seats:

- seat 1
- seat n (last passenger)

Every time someone is forced to pick randomly, the choice is only between the first and last seats eventually.

Thus:

$$\mathbb{P}(\text{last person gets own seat}) = \frac{1}{2}$$

Problem 1.2.3 Random Meeting Problem

Two friends agree to meet between 0 and 1 hour. Each arrives at a random time uniformly and waits 15 minutes. What is the probability they meet?

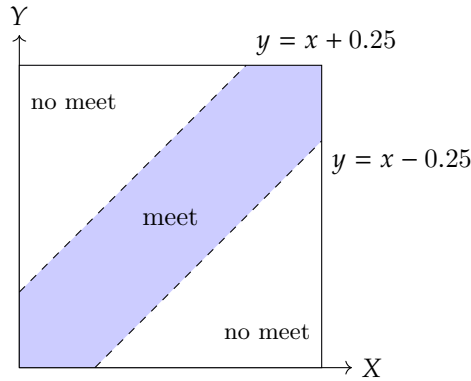
Solution: Solution via geometry

You meet if:

$$|X - Y| < 0.25$$

In the unit square, the “meeting band” is:

$$\begin{aligned}
 \text{Area} &= 1 - 2 \times (\text{triangles each with side } 0.75) \\
 &= 1 - 2 \times \frac{1}{2}(0.75)^2 \\
 &= 1 - 0.5625 \\
 &= 0.4375.
 \end{aligned}$$



Problem 1.2.4 Harder Version — Different Wait Times

Two friends agree to meet between time 0 and 1 hour.

- Friend A arrives uniformly on $[0, 1]$ and waits a hours
- Friend B arrives uniformly on $[0, 1]$ and waits b hours

(where $0 < a, b < 1$.)

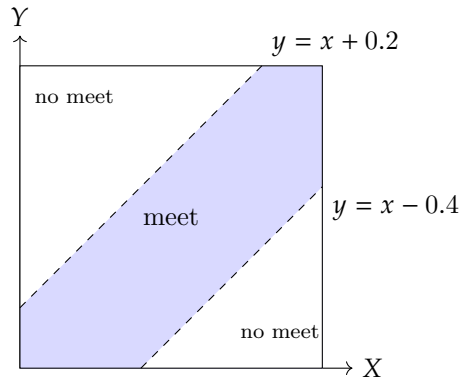
What is the probability they meet?

Solution:

Two friends arrive at times X and Y on $[0, 1]$. Friend A waits a hours and B waits b hours. They meet exactly when the arrival intervals overlap, which is $-b \leq Y - X \leq a$. For uniform arrivals this is the area of a diagonal band in the unit square. You remove two no-meeting triangles of area $\frac{1}{2}(1-a)^2$ and $\frac{1}{2}(1-b)^2$, so

$$\mathbf{P}(\text{meet}) = 1 - \frac{(1-a)^2 + (1-b)^2}{2}.$$

For arbitrary arrival distributions you integrate the joint density over that band.



Problem 1.2.5 The Birthday Paradox

How many people are needed so that the probability of at least one shared birthday is 50%?

Solution:

$$\mathbb{P}(\text{no match}) = \frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{365 - n + 1}{365}$$

Set equal to $\frac{1}{2}$, solve numerically.

Problem 1.2.6 The Balls & Bins Problem

Throw n balls into n bins independently and uniformly. What is the expected number of empty bins?

Solution: For a given bin:

$$\mathbb{P}(\text{empty}) = \left(1 - \frac{1}{n}\right)^n \approx e^{-1}$$

So:

$$\mathbb{E}[\text{empty bins}] = ne^{-1}$$

Deep Dive: Gamma(Shape, Scale)

Gamma distributions show up in:

- waiting times
- variance-gamma models (used in quant finance)
- Bayesian statistics
- mixture models

Properties

$$X \sim \Gamma(k, \theta)$$

$$\mathbb{E}[X] = k\theta, \quad \text{Var}(X) = k\theta^2$$

Interviews often test “identify the right distribution” when:

- time between jumps
- sum of exponential variables

Quant Interview Questions

Problem 1.2.7

Roll a die repeatedly. What is the expected number of rolls until you see the sequence 1-1 (two ones in a row)?

Solution:

This is a Markov chain with 2 states:

S_0 = no previous 1

S_1 = last roll was 1

Solve linear equations: Expected rolls = $36/11 \approx 3.27$

Problem 1.2.8

What is the probability that the maximum of two independent Uniform(0,1) variables exceeds 0.8?

Solution:

$$1 - 0.8^2 = 0.36$$

Problem 1.2.9

A fair coin is tossed until heads. Expected tosses?

Solution:

$$\frac{1}{p} = 2.$$

Problem 1.2.10

A box has 3 white and 2 black balls. Two are drawn without replacement. Probability both white?

Solution:

$$\frac{3}{5} \cdot \frac{2}{4} = 0.3.$$

Problem 1.2.11

What is the expected value of the maximum of 3 IID Uniform(0,1) variables?

Solution:

$$\frac{3}{4}$$

Problem 1.2.12

What is $\mathbb{E}[X|X > 1]$ for exponential rate 1?

Solution:

$$\mathbb{E}[X - 1|X > 1] + 1 = 2.$$

Problem 1.2.13

Draw a random point in a unit square. Probability $x + y < 1$?

Solution:

$$= \text{area of right triangle} = 1/2.$$

Problem 1.2.14

Flip a fair coin 10 times. Probability of exactly 5 heads?

Solution:

$$\binom{10}{5}/2^{10}$$

Problem 1.2.15

If two stocks have correlation ρ , what is the covariance?

Solution:

$$\text{Cov} = \rho\sigma_1\sigma_2$$

Problem 1.2.16

Probability that Poisson(λ) variable is even?

Solution:

$$\frac{1}{2}(1 + e^{-2\lambda})$$

Brainteasers**Brainteasers 1.2.1**

The Light Switch Puzzle (Short Version)

Solution:

Turn switch A on for 5 minutes

Turn off A, turn on B

Enter room:

Light on $\rightarrow$ B

Light off but warm $\rightarrow$ A

Light off and cold $\rightarrow$ C

Brainteasers 1.2.2

Two Ropes Burning (45 Minutes)

Solution:

Burn Rope 1 at both ends

Burn Rope 2 at one end

When Rope 1 finishes (30 min), light Rope 2 other end

Rope 2 burns remaining 15 min

Total = 45 min

Brainteasers 1.2.3

100 Lockers

Solution:

Only lockers with square numbers stay open (because toggled odd number of times).

Brainteasers 1.2.4

Find Heavier Ball With Two Weighings

Solution:

Use a $3 \times 3 \times 3$ splitting; identify remainder group recursively.

Brainteasers 1.2.5

Farmer, Wolf, Sheep, Cabbage

Solution: Sheep first, return alone, wolf next, bring back sheep, cabbage over, return empty, sheep last.

Brainteasers 1.2.6

Mislabeled Bags

Solution: Draw one fruit from the “mixed” label (which must be wrong). Deduce all labels.

Brainteasers 1.2.7

Ants on a Stick

Solution: Expected time until fall depends on boundaries and speeds— but direction doesn’t matter: ants can be treated as ghosts passing through.

Brainteasers 1.2.8

12 Coins, One Fake (Unknown Heavier/Light)

Solution: Three weighings $\rightarrow$ ternary decision tree ($27 \text{ leaves} \geq 24 \text{ possibilities}$).

Brainteasers 1.2.9

The Hotel Bill Problem

Solution: (paradox with missing \$1; resolve misdirection)

Brainteasers 1.2.10

Monty Hall Variation

Solution: Stay vs switch when multiple goats opened $\rightarrow$ generalize to $(n - 1)/n$ win rate when switching.

1.3 Advanced Probability, Order Statistics, More Interview Questions, and More Brainteasers

Advanced Probability: Decomposition, Law of Total Expectation

Quant interview questions often require breaking a complex expectation into simpler conditional pieces. The law of total expectation:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$$

This is especially useful when:

- distributions change after events
- we condition on discrete states
- multiple processes interact

Example 1.3.1 (Expected number of rolls to get a 6 or double 1s)

Break into states:

- before starting
- after rolling a 1
- after partial matching
- absorbing states

Recursion becomes easier than direct computation.

Advanced Bayes Rule Use-Cases in Interviews

Interviewers often hide Bayes' rule implicitly. Key patterns:

1. A result is observed from a mixture distribution
2. The data come from multiple underlying states
3. Need to invert probability to infer the state

Canonical pattern: Signal extraction under uncertainty.

Example 1.3.2

A stock is predicted by two analysts, one reliable (80%), one unreliable (50%). A buy signal appears. What's the probability the stock indeed goes up?

This is pure Bayes:

$$P() = \frac{P(\text{Signal} | U_p)P(U_p)}{P(\text{Signal})}$$

This structure appears in:

- credit rating transitions
- signal-based trading
- Bayesian volatility estimation
- model uncertainty

Order Statistics (Deep Dive)

Order statistics are among the least studied but most tested interview topics.

Given $X_1, \dots, X_n$ i.i.d. with CDF $F(x)$ and PDF $f(x)$:

- minimum: $X_{(1)}$
- maximum: $X_{(n)}$
- k-th order statistic: $X_{(k)}$

General formula

The probability density function (PDF) of the k -th order statistic, denoted as $X_{(k)}$, for n i.i.d. random variables is given by:

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1-F(x)]^{n-k} f(x)$$

Where: $f(x)$ is the PDF of the population. $F(x)$ is the Cumulative Distribution Function (CDF) of the population.

Order Statistics for n Uniform(0, 1) Variables

When the underlying distribution is $U(0, 1)$, we have $f(x) = 1$ and $F(x) = x$ for $x \in [0, 1]$.

Expected Maximum: The expected value of the largest observation $X_{(n)}$ is:

$$\mathbb{E}[X_{(n)}] = \frac{n}{n+1}$$

Proof. Find the CDF: $F_{X_{(n)}}(x) = P(X_1 \leq x, \dots, X_n \leq x) = [F(x)]^n = x^n$ Find the PDF: $f_{X_{(n)}}(x) = \frac{d}{dx}x^n = nx^{n-1}$ Calculate Expectation:

$$\mathbb{E}[X_{(n)}] = \int_0^1 x \cdot nx^{n-1} dx = \int_0^1 nx^n dx = \frac{n}{n+1}$$

⊕

Expected Minimum: The expected value of the smallest observation $X_{(1)}$ is:

$$\mathbb{E}[X_{(1)}] = \frac{1}{n+1}$$

Expected Range: The range is defined as the difference between the maximum and the minimum ($X_{(n)} - X_{(1)}$). By the linearity of expectation:

$$\mathbb{E}[\max - \min] = \mathbb{E}[X_{(n)}] - \mathbb{E}[X_{(1)}]$$

$$\mathbb{E}[\max - \min] = \frac{n}{n+1} - \frac{1}{n+1} = \frac{n-1}{n+1}$$

Note: A helpful intuition for the $U(0, 1)$ case is that n points partition the interval $[0, 1]$ into $n+1$ equal segments on average. Each segment has an expected length of $\frac{1}{n+1}$. The minimum is the first segment: $\frac{1}{n+1}$. The maximum is n segments: $\frac{n}{n+1}$.

Advanced Interview Questions (20 More)

Problem 1.3.1

You draw two independent exponential(1) variables. What is the probability their difference exceeds 1?

Solution:

Let $X, Y \sim \text{Exp}(1)$. We want to compute:

$$\mathbb{P}(X - Y > 1) = \int_1^\infty \int_0^{x-1} e^{-x} e^{-y} dy dx$$

First, we evaluate the inner integral with respect to y :

$$= \int_1^\infty e^{-x} (1 - e^{-(x-1)}) dx$$

Next, we distribute the e^{-x} term to simplify the integrand:

$$= \int_1^\infty e^{-x} dx - \int_1^\infty e^{-2x+1} dx$$

Final answer Solving the definite integrals:

$$= e^{-1} - \frac{1}{2}e^{-1} = \frac{1}{2e}$$

Problem 1.3.2

Let X Uniform(0,2). Compute $E[X - X_i | 1]$.

Solution:

Conditional distribution is uniform on (1,2).

$$E[X|X > 1] = 1.5$$

Problem 1.3.3

A coin gives heads with probability p . Expected number of HTH pattern occurrences in n flips?

Solution:

$$(n-2)p^2(1-p)$$

Problem 1.3.4

What is the expected number of fixed points in a random permutation of n elements?

Solution:

$$1$$

Reason: indicator variable trick.

Problem 1.3.5

Two stocks have correlation ρ . What is the distribution of their sum?

Solution:

if:

$$X, Y \sim N(\mu, \sigma^2), \quad \text{Corr} = \rho$$

Then;

$$X + Y \sim N(2\mu, 2\sigma^2 + 2\rho\sigma^2)$$

Problem 1.3.6**Problem 1.3.7**

How do you simulate correlated normals?

Solution:

Cholesky:

Let

$$\Sigma = LL^T$$

Then:

$$X = LZ, \quad Z \sim N(0, I)$$

Problem 1.3.8

Poisson arrivals with rate λ . What is the expected time until the 3rd arrival?

Solution:

Sum of three exponentials:

Gamma($3, \frac{1}{\lambda}$)

$$\mathbb{E} = \frac{3}{\lambda}$$

Problem 1.3.9

You draw 10 balls from an urn with replacement, each red with p . Distribution of total reds?

Solution:

Binomial($10, p$)

Problem 1.3.10

Flip coin until 2 heads occur; expected flips?

Solution:

Negative binomial:

$$\frac{2}{p}$$

Problem 1.3.11

Probability that two independent random angles differ by more than 90 degrees?

Solution:

$$\frac{1}{2} \quad (\text{symmetry}).$$

Problem 1.3.12

What is the variance of a geometric(p) variable (counting trials until first success)?

Solution:

$$\text{Var} = \frac{1-p}{p^2}$$

Problem 1.3.13

$X \sim \text{Exp}(\lambda)$. Distribution of $\min(X, c)$?

Solution:

$$\mathbb{P}(\min(X, c) > t) = e^{-\lambda t}, \quad t < c$$

Mass at c :

$$e^{-\lambda c}$$

Problem 1.3.14

Let X and Y be independent normals. What's the distribution of $X - Y$?

Solution:

Still normal,

$$\begin{aligned}\text{Mean} &= \mu_1 - \mu_2 \\ \text{Variance} &= \sigma_1^2 - \sigma_2^2\end{aligned}$$

Problem 1.3.15

For IID variables, what is $P(\max < a)$?

Solution:

$$F(a)^n$$

Problem 1.3.16

Consider symmetric random walk. What is probability of ever hitting +1?

Solution:

Because 1D symmetric walk is recurrent.

Problem 1.3.17

Expected number of uniform(0, 1) draws until sum exceeds 1?

Solution:

$$e \approx 2.718$$

Classic surprising result.

Problem 1.3.18

What is $E[\min(X, Y)]$ for IID exponential(1)?

Solution:

$$E[\min] = \frac{1}{2}$$

Problem 1.3.19

Draw a random chord in a circle. What is the probability its length exceeds the radius?

Solution:

Problem 1.3.20

Probability that two random points in unit square are within distance $< r$?

Solution:

Problem 1.3.21

Roll until total score exceeds 12. Expected rolls?

Solution:

Linear recursion $\rightarrow \sim 4.2$ rolls.

Brainteasers (10 More, With Solutions)

Brainteasers 1.3.1 The Poison Wine Problem (Classic)

1000 bottles, 1 is poisoned, death occurs in 24 hours. You have 10 prisoners to test. How do you identify bottle?

Binary encoding $\rightarrow$ 10 prisoners can identify $2^{10} = 1024$ bottles.

Brainteasers 1.3.2 The Random Ants on a Triangle Problem

Expected collision events depend only on direction choices. But whether they collide or not = same as asking if two choose opposite directions.

Brainteasers 1.3.3 Man Finds 3 Chest Keys, Only One Works

Expected attempts = 2 (because correct key equally likely among remaining after each failure).

Brainteasers 1.3.4 100 Prisoners: One Light Bulb Puzzle

Use designated "counter". Solution requires synchronization of events. Interviewers test ability to design distributed algorithms.

Brainteasers 1.3.5 Toggle Bits Problem

Given n bits initially 0, toggle bits based on divisors $\rightarrow$ bit i ends up toggled $d(i)$ times. Prime if toggled twice, perfect squares toggled odd times, etc.

Brainteasers 1.3.6 'Truth or Lies' Problem

If one person always lies and one always tells truth, use symmetric questions like: "If I asked the other person which door leads to freedom, what would he say?"

Brainteasers 1.3.7 100 Coins, Blindfolded, Split into 2 piles with equal # of heads

Flip any chosen coins in second pile. Uses invariance.

Brainteasers 1.3.8 3 Switching Ladders Puzzle

Cycle detection problem $\rightarrow$ converts to graph theory.

Brainteasers 1.3.9 Two Glasses, 100-floor Building

Minimum worst-case drops = 14 (because $14 + 13 + \dots + 1 = 105 \geq 100$).

Brainteasers 1.3.10 Train and Fly Puzzle

Two trains move toward each other, fly goes back and forth at speed v . Distance = $v \cdot$ total time until collision.

Interviewers test recognition of geometric series vs simple physical reasoning.

1.4 Brownian Motion, Scaling Limits, Quadratic Variation, Hitting Times, Jensen's Inequality

Crucial Property: Quadratic Variation

Stopping Times - Deepening the Intuition

Hitting Times of Brownian Motion

Jensen's Inequality (Very Important)

Optional Stopping Theorem (Intuition Only)

1.5 Statistics, Estimation & Sampling Theory

Method of Moments (MoM)

Maximum Likelihood Estimation (MLE)

Large Sample (Asymptotic) Theory

Covariance Matrices & Portfolio Statistics

1.6 Brainteasers & Quant Puzzles

Multicollinearity

Heteroskedasticity and White Standard Errors

Autocorrelation & HAC / Newey-West Standard Errors

Logistic Regression

1.7 Time-Series Models, Forecasting, Stationarity, Cointegration, Kalman Filter, GARCH

Random Walks (Very Common in Finance)

Vector Autoregressions (VAR)

State-Space Models & Kalman Filter

1.8 Machine Learning for Quants

Supervised Learning Overview

Support Vector Machines (SVM)

K-Nearest Neighbors (kNN)

Clustering Methods: K-Means, Hierarchical, SOM

Principal Component Analysis (PCA)

Neural Networks: Foundation for LSTM & Deep Learning

Feature Engineering

Chapter 2

Derivatives, Greeks, Models & Stochastic Calculus

2.1 Black-Scholes-Merton Derivations

2.2 Greeks

2.3 Stochastic calculus

2.4 Hedging & replication

Chapter 3

Coding, Math, Linear Algebra & Technical Computing

3.1 Coding Fundamentals for Quants (Python + C++)

3.2 Math Essentials

3.3 Practical Quant Engineering

Chapter 4

Numerical Methods, C++, & HFT-Style Algorithms

- 4.1 Numerical Methods in Quant Finance
- 4.2 C++ for Quants (Low-Latency Coding)
- 4.3 High-Frequency Trading (HFT) Algorithms & Data Structures
- 4.4 Numerical Option Pricing
- 4.5 Probabilistic & Statistical Tools for HFT

Chapter 5

Portfolio Optimization, Execution, & Systematic Strategy Design

5.1 Signal Construction & Alpha Modeling

5.2 Position Sizing & Risk Constraints

5.3 Statistical Arbitrage & Pairs Trading

5.4 Backtesting & Simulation Frameworks

5.5 Execution Algorithms & Cost Modeling